

Supplementary Information

Terrain-Conditioned Cross-Attention for Cross-Region Avalanche Debris Mapping with Sentinel-1 SAR

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S1. Radar-topographic formulation

For terrain slope θ , aspect α , and local incidence angle, the terrain tensor contains $\cos(\text{LIA})$, scaled slope, $\sin \alpha$, $\cos \alpha$, scaled elevation, and the fixed-reference coordinate $\theta_{78} = \cos(\alpha - 78^\circ)$. The final channel is an aspect coordinate relative to a fixed nominal ascending reference, not an authoritative per-event radar heading, and it does not make SAR responses physically invariant. AvalCD supplies cross/co channel roles as VH/VV for Livigno, Tromsø, and Pish and HV/HH for Nuuk. The loader uses supplied decibel rasters without percentile scaling: values outside $[-45, 15]$ dB are marked invalid, then missing cross/co values are replaced by $-20/-12$ dB after valid-mask construction. A shared encoder forms multiscale representations, and each Radar-Topographic Cross-Attention Block uses dense SAR-change queries and an 8×8 pooled terrain context.

The attended context O_l and spatial terrain gate are

$$O_l = \text{softmax} \left(Q_l K_l^\top / \sqrt{d} \right) V_l, \quad (1)$$

$$W_l = \sigma \{ \text{Conv}_{1 \times 1} [\text{GELU}(\text{Conv}_{3 \times 3}(T_l))] \}, \quad (2)$$

$$\tilde{F}_l = \text{BatchNorm}(F_l^\Delta + O_l) \odot (1 + W_l). \quad (3)$$

The loss is positive-weighted binary cross-entropy plus Dice loss and a slope-prior term weighted by 0.5. The slope prior acts below 5° and above 65° . It is not a validity mask because genuine labels occupy those strata.

S2. Training, inference, and statistical protocol

Training uses 128×128 patches with 64-pixel stride and a 1:1 positive/negative patch ratio, yielding 2,054 patches per epoch. All methods use AdamW, initial learning rate 2×10^{-4} , cosine decay, six epochs, batch size 16, weight decay 10^{-4} , and gradient clipping at 5. Checkpoints maximize balanced validation-patch F1 at threshold 0.5. A full-scene threshold is then selected only on the two validation rasters over 0.20–0.80 in 0.02 increments. Frozen thresholds are applied to Tromsø and Pish.

Checkpoint-era augmentation reflects aspect sine under horizontal flips and aspect cosine under vertical flips. Spatial arrays are rotated orthogonally, but the precomputed fixed-reference aspect channel is not directionally recomputed. The released training path is therefore not fully equivariant; directional analyses are descriptive.

Full-scene inference uses Hanning blending. Pixel metrics use valid-mask pixels only. Instance hit rate requires overlap of at least 30% or 50% of the reference polygon. Statistical inference includes 1,000-draw paired spatial-block bootstraps and paired t -tests over three training seeds.

Spatial blocks quantify map-sampling variability conditional on a checkpoint; they do not replace training-seed uncertainty.

S3. Complete spatial and seed-level comparisons

All reported three-seed mean $\pm$ SD entries use descriptive population standard deviation (ddof = 0) over the fixed seeds 42, 123, and 456. Paired t -test intervals separately use sample variance.

Table 1: Paired comparisons of TopoRadar-Net with baselines. Parts A and B use Seed 42 spatial blocks; Part C uses pooled macro F1 over three training seeds

Analysis	Comparator	Δ F1	95% CI	p
Tromsø, 143 blocks	SiamUNet-diff	+6.39	[+2.59, +10.59]	< 0.001
	ResU-Net	+4.35	[+1.33, +7.38]	0.010
	Self-attention U-Net	+2.30	[+0.27, +4.41]	0.030
	SiamUNet-conc	-1.12	[-2.80, +0.40]	0.144
Two regions, 339 blocks	SiamUNet-diff	+12.77	[+9.18, +16.61]	< 0.001
	ResU-Net	+4.30	[+1.54, +6.97]	0.002
	Self-attention U-Net	+3.27	[+1.12, +5.37]	0.010
	SiamUNet-conc	+2.47	[+0.41, +4.60]	0.026
Three seeds, macro F1	SiamUNet-diff	+8.92	[+3.25, +14.59]	0.021
	ResU-Net	+0.86	[-3.23, +4.96]	0.459
	Self-attention U-Net	+0.38	[-8.28, +9.04]	0.869
	SiamUNet-conc	+0.65	[-4.69, +5.99]	0.651

Repeating the pooled 339-block bootstrap separately for seeds 42, 123, and 456 yields +3.26 percentage points (95% CI +1.32 to +5.33, $p = 0.002$), -2.99 (-6.67 to +0.08, $p = 0.060$), and +4.16 (+1.00 to +7.27, $p = 0.016$) relative to the self-attention U-Net. The descriptive mean is $+1.48 \pm 3.18$ percentage points and is positive in two of three seeds.

S4. Instance completeness

Table 2: Tromsø polygon hit rates across three seeds. D1–D4 are European Avalanche Warning Services size classes

Method	HR@0.3	HR@0.5	D1	D2	D3	D4
SiamUNet-diff	63.25 $\pm$ 2.52	54.42 $\pm$ 2.80	6.7	33.3	71.8	89.6
SiamUNet-conc	82.62 $\pm$ 1.07	74.07 $\pm$ 1.40	13.3	61.3	91.1	100.0
ResU-Net	76.92 $\pm$ 4.83	68.09 $\pm$ 4.20	6.7	53.3	85.9	95.8
Self-attention U-Net	79.20 $\pm$ 2.45	69.52 $\pm$ 2.10	6.7	58.7	86.9	100.0
TopoRadar-Net	75.50 $\pm$ 1.61	68.09 $\pm$ 1.95	20.0	49.3	84.0	95.8

S5. Terrain-stratum sensitivity

Across three seeds, backslope differences average -0.34 ± 6.28 points in Tromsø and $+1.20 \pm 7.73$ in Pish. Tromsø foreslopes average -1.21 ± 0.86 ; Pish runout/valley terrain averages $+5.32 \pm 7.76$. Extreme-steep strata are highly seed-variable. These data do not support a stable component-isolated directional mechanism.

Table 3: Seed-42 F1 by terrain stratum. The final column is TopoRadar-Net minus the self-attention U-Net

Region	Stratum	TopoRadar	Self-attention U-Net	Δ F1
Tromsø	Foreslope	79.94	80.01	-0.07
	Cross-slope	68.93	69.43	-0.50
	Backslope	81.08	75.23	+5.85
	Runout/valley	76.76	77.91	-1.15
	Avalanche chute	81.34	77.00	+4.34
	Extreme steep	63.37	54.23	+9.14
Pish	Foreslope	51.27	50.13	+1.14
	Cross-slope	46.96	46.92	+0.03
	Backslope	53.42	48.33	+5.09
	Runout/valley	45.97	38.22	+7.75
	Avalanche chute	55.82	56.25	-0.44
	Extreme steep	8.85	16.65	-7.80

S6. Validation-scene comparison

Table 4: Performance on the Nuuk 11 April 2021 validation scene across three seeds

Method	F1	IoU	Precision	Recall
SiamUNet-diff	59.35 ± 1.58	42.22 ± 1.45	58.10 ± 0.58	60.79 ± 3.70
SiamUNet-conc	66.53 ± 0.97	49.85 ± 1.09	57.72 ± 2.15	78.66 ± 1.74
ResU-Net	67.16 ± 1.62	50.57 ± 1.81	60.10 ± 4.51	76.69 ± 3.48
Self-attention U-Net	67.01 ± 1.98	50.43 ± 2.22	57.06 ± 2.94	81.31 ± 0.51
TopoRadar-Net	66.75 ± 1.96	50.13 ± 2.22	56.63 ± 3.31	81.50 ± 1.10

S7. Construct and operational audits

The slope-prior mask overlaps 19,663 of 106,442 training positives (18.47%), 824 of 86,802 validation positives (0.95%), and 492 of 56,548 held-out positives (0.87%). Three-seed recall inside the held-out penalty stratum is $50.47 \pm 3.18\%$ for the full model versus $71.29 \pm 6.39\%$ for No-GeoLoss in Tromsø, and $1.08 \pm 1.52\%$ versus $6.45 \pm 6.97\%$ in Pish.

For the corridor proxy, Platt scaling improves Brier score from 0.282 to 0.208 in Tromsø and from 0.462 to 0.198 in Pish. TopoRadar-Net Pish alerts have $83.0 \pm 2.3\%$ precision and $62.5 \pm 10.2\%$ recall; the self-attention U-Net has $77.4 \pm 6.1\%$ and $70.8 \pm 5.9\%$. Both miss the only Tromsø road-intersecting reference component. Cached full-scene processing is below 5 s per scene on Apple MPS, excluding acquisition and human review.

S8. Reproducibility identifiers

The publication archive contains three seeds for TopoRadar-Net, the bottleneck self-attention U-Net (stored under the historical artifact key “Attention U-Net”), and No-GeoLoss (nine checkpoints). The verification suite checks stored-summary arithmetic, parameter counts, valid-mask confusion closure, physical-area denominators, calibration bounds, and construct-audit values; raw-raster re-training is outside the package gate. Persistent identifying repository links are withheld in this review copy to preserve double blindness and are provided to the editorial office.